

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
QUEUE_RX	READY
QUEUE_TX	READY
LED	READY
SEM_RX	BLOCKED
SEM_TX	BLOCKED
MUTEX1	BLOCKED
MUTEX2	BLOCKED
MEMORY	BLOCKED
SENSOR	BLOCKED
UART	BLOCKED

```
-----
[QUEUE] Sent: 8
[LED] Task Running
[SEM] Receiver Received Signal
[QUEUE] Received: 8
[UART] Task Running
[SENSOR] Task Running
[MUTEX] Task2 Acquired
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[LED] Task Running
[QUEUE] Sent: 9
[APP] Task Running
```

```
=====
MiniRTOS
ARM Cortex-M4 RTOS
=====
```

System Initialization...

```
[OK] UART Driver
[OK] SysTick Timer
[OK] Scheduler
[OK] Memory Manager
[OK] IPC Objects
```

```
=====
MiniRTOS Boot
=====
=====
```

```
Tasks Created Successfully
Starting Scheduler...
=====
```

```
[UART] Task Running
[SENSOR] Task Running
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
MEMORY	READY
SEM_RX	READY
SEM_TX	READY
MUTEX1	READY
MUTEX2	READY
QUEUE_TX	READY
QUEUE_RX	READY
LED	READY
SENSOR	BLOCKED
UART	BLOCKED

```
-----
[MEMORY] Task Running
[SEM] Sender Giving Signal
[MUTEX] Task1 Locked
[QUEUE] Sent: 0
[QUEUE] Received: 0
[LED] Task Running
[SEM] Receiver Received Signal
[MUTEX] Task1 Released
[MUTEX] Task2 Waiting
[SEM] Receiver Received Signal
[SENSOR] Task Running
[UART] Task Running
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
UART	BLOCKED
SENSOR	BLOCKED
SEM_RX	BLOCKED
QUEUE_RX	BLOCKED
MUTEX2	BLOCKED
MUTEX1	BLOCKED
LED	BLOCKED
QUEUE_TX	BLOCKED
SEM_TX	BLOCKED
MEMORY	BLOCKED

```
-----
[SEM] Sender Giving Signal
[LED] Task Running
[QUEUE] Sent: 1
[SEM] Receiver Received Signal
[MUTEX] Task2 Acquired
[QUEUE] Received: 1
[UART] Task Running
[SENSOR] Task Running
[MEMORY] Task Running
[APP] Task Running
=====
```

```
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MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
SEM_TX	READY
MEMORY	BLOCKED
SENSOR	BLOCKED
UART	BLOCKED
SEM_RX	BLOCKED
QUEUE_RX	BLOCKED
MUTEX1	BLOCKED
MUTEX2	BLOCKED
QUEUE_TX	BLOCKED
LED	BLOCKED

```
-----
[SEM] Sender Giving Signal
[MUTEX] Task2 Released
[LED] Task Running
[QUEUE] Sent: 2
[SEM] Receiver Received Signal
[QUEUE] Received: 2
[SEM] Receiver Received Signal
[UART] Task Running
[SENSOR] Task Running
[APP] Task Running
=====
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
SENSOR	BLOCKED
UART	BLOCKED
QUEUE_RX	BLOCKED
SEM_RX	BLOCKED
MUTEX1	BLOCKED
QUEUE_TX	BLOCKED
LED	BLOCKED
MUTEX2	BLOCKED
SEM_TX	BLOCKED
MEMORY	BLOCKED

```
-----
[MUTEX] Task2 Acquired
[SEM] Sender Giving Signal
[QUEUE] Sent: 3
[LED] Task Running
[SEM] Receiver Received Signal
[MUTEX] Task2 Released
[QUEUE] Received: 3
[UART] Task Running
[SENSOR] Task Running
[MEMORY] Task Running
[APP] Task Running
```

```
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MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
MEMORY	BLOCKED
SENSOR	BLOCKED
UART	BLOCKED
SEM_RX	BLOCKED
QUEUE_RX	BLOCKED
MUTEX1	BLOCKED
MUTEX2	BLOCKED
LED	BLOCKED
QUEUE_TX	BLOCKED
SEM_TX	BLOCKED

```
-----
[QUEUE] Sent: 4
[SEM] Sender Giving Signal
[LED] Task Running
[SEM] Receiver Received Signal
[QUEUE] Received: 4
[MUTEX] Task1 Locked
[SEM] Receiver Received Signal
[UART] Task Running
[SENSOR] Task Running
[MUTEX] Task1 Released
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
MUTEX2	READY
SEM_TX	READY
QUEUE_TX	READY
MUTEX1	BLOCKED
SENSOR	BLOCKED
UART	BLOCKED
SEM_RX	BLOCKED
QUEUE_RX	BLOCKED
LED	BLOCKED
MEMORY	BLOCKED

```
-----
[MUTEX] Task2 Waiting
[SEM] Sender Giving Signal
[QUEUE] Sent: 5
[SEM] Receiver Received Signal
[LED] Task Running
[QUEUE] Received: 5
[UART] Task Running
[SENSOR] Task Running
[MEMORY] Task Running
[MUTEX] Task1 Locked
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
MUTEX2	READY
MUTEX1	BLOCKED
MEMORY	BLOCKED
SENSOR	BLOCKED
UART	BLOCKED
QUEUE_RX	BLOCKED
SEM_RX	BLOCKED
LED	BLOCKED
QUEUE_TX	BLOCKED
SEM_TX	BLOCKED

```
-----
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[LED] Task Running
[QUEUE] Sent: 6
[QUEUE] Received: 6
[MUTEX] Task1 Released
[MUTEX] Task2 Waiting
[SEM] Receiver Received Signal
[UART] Task Running
[SENSOR] Task Running
[SEM] Sender Giving Signal
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	STATE
APP	RUNNING
IDLE	READY
SEM_TX	BLOCKED
SENSOR	BLOCKED
UART	BLOCKED
QUEUE_RX	BLOCKED
SEM_RX	BLOCKED
MUTEX2	BLOCKED
MUTEX1	BLOCKED
QUEUE_TX	BLOCKED
LED	BLOCKED
MEMORY	BLOCKED

```
=====
[SEM] Receiver Received Signal
[LED] Task Running
[QUEUE] Sent: 7
[MUTEX] Task2 Acquired
[QUEUE] Received: 7
[UART] Task Running
[SENSOR] Task Running
[MEMORY] Task Running
[MUTEX] Task2 Released
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[APP] Task Running
```